Title 0:00–0:15. Joint work with Ken Brown.

0:15–2:15 / 1 Goal and result Need T or CCZ. Distillation overhead. One $T/T^\dagger$ layer as CCZ. $[\![8,3,2]\!]$: d_Z=2, run on hardware [MRV24]. Q48: $(16,3)$, $26T + 22T^\dagger$.

CSS codes and transversal gates

2:15–2:45 / 2 Outline Seven parts. The last is Q48 in a CCZ factory.

2:45–3:45 / 3 Generator rows Columns labelled by cube points. Rows u_1 = 00001111, u_2. Overlap 00000011 is an edge. XOR.

3:45–4:30 / 4 Pauli strings $X(u_1)$ on a face. Overlap 2 commutes, overlap 1 anticommutes.

4:30–5:30 / 5 Stabilizer, CSS Syndrome, $[\![n,k,d]\!]$. $X(a)$ for $a \in C_2$, $Z(b)$ for $b \in C_1^\perp$. $G = [K;S]$, $|x\rangle_L$.

5:30–7:00 / 6 $[\![8,3,2]\!]$ code Rows $K_i = u_i$, $S = \mathbf{1}$. Pairs 00001111, 11110000, $x = 111$. Z on 000: syndrome 1. Edge: logical, d_Z=2, d_X=4.

7:00–8:15 / 7 Clifford, non-Clifford Gates on basis states. CCZ flips $|111\rangle$. Hierarchy: T, CCZ in level 3. Add T or CCZ for universality.

8:15–9:00 / 8 Transversal, distillation One gate per qubit. Eastin–Knill: not universal. Distillation needs transversal T or CCZ.

9:00–10:05 / 9 Three codes Steane: H, S. 15-qubit: $T^{\otimes 15} = \bar{T}^\dagger$. $[\![8,3,2]\!]$: T/T† gives CCZ. State the question.

$[\![8,3,2]\!]$ **example** *10:05–10:15: outline, current theme highlighted.*

10:15–11:15 / 10 Signed weights Integer signed weight, mod 8. Every codeword in a logical coset needs the same phase.

11:15–13:15 / 11 Pattern from overlaps Ask where $T^\dagger$ goes. Reveal $\gamma = u_1 + u_2 + u_3$. Face 0, edge 0, vertex -1.

13:15–14:00 / 12 Row congruences Coefficients 1, -2, 4 give mod 8, 4, 2. Only the logical triple is odd.

Gate conditions *14:00–14:10: outline, current theme highlighted.*

14:10–15:10 / 13 Triorthogonal codes Timeline. Triorthogonal: even overlaps, but CCZ needs one odd triple.

15:10–16:55 / 14 Nine CH conditions Logical, mixed, stabilizer families. K3 odd. Correction may contain CZ. $T/T^\dagger$: $S^\dagger$ on the $T^\dagger$ qubits.

Geometric construction *16:55–17:05: outline, current theme highlighted.*

17:05–18:25 / 15 Q48 result 48 physical, 3 logical. d_X=16, d_Z=3. 26 T and 22 inverse, no correction.

18:25–19:25 / 16 Fix the stabilizers Projective columns. 8-divisible span. Find K. Solve signs. Divisibility is a choice.

19:25–20:25 / 17 Projective syndromes Nonzero columns detect one Z. Distinct columns detect two. This proves only d_Z.

20:25–21:25 / 18 48-point support 63 minus 15 gives 48. Remove w = 00. Three blocks of 16. Six coordinate rows.

21:25–22:55 / 19 Weights 24 and 32 Nonzero u coefficient: 3 times 8, 60 codewords. Pure w coefficient: 2 times 16, 3 codewords.

22:55–23:55 / 20 Geometric family Recover 48 at $p = 4, q = 2$. Other counts are lower bounds. Star is a fixed-support census.

Column picture and completion *23:55–24:05: outline, current theme highlighted.*

24:05–25:25 / 21 Column indicators Occupied columns as a Boolean function. $[\![8,3,2]\!]$: y. Q48: $w_1 + w_2 + w_1w_2$.

25:25–26:55 / 22 Moments and monomials Overlap parity is a monomial moment. One exceptional cubic. State the coset briefly.

26:55–28:25 / 23 Logical labels Three label bits per point. Product forces $x = \kappa$. S-only constraints already hold.

28:25–30:25 / 24 Logical completion One-row space, compatible pairs, linear third row. Three-site witness makes d_Z=3.

30:25–31:55 / 25 $T/T^\dagger$ signs Binary gamma in mod-4 equations. Show the polynomial, count $6 + 6 + 10$ inverse gates.

Lower bounds *31:55–32:05: outline, current theme highlighted.*

32:05–33:55 / 26 Reed–Muller supports Positive-degree tests vanish at 0. Correct weight parity, then use RM duality.

33:55–35:25 / 27 Exclusion through 38 Rank at most 8, support reductions, complete enumeration, no logical completion.

35:25–37:25 / 28 Length comparison Our 38 floor. WJS 39–46. 47 quasi-transversal. Sign failure is code-specific.

CCZ factory and questions *37:25–37:35: outline, current theme highlighted.*

37:35–39:35 / 29 Distillation Encode $|{+}{+}{+}\rangle$, 48 noisy T, measure, decode. $d_Z \geq 3$. JVB 47-to-1, footprint 9, CZ correction. n is not footprint.

39:35–41:05 / 30 Known and open Ranks 6 and 9, weights 24 and 32, 39 Z generators, logicals 16, 16, 18. Open: preparation, noise, costs, bias.

41:05–42:35 / 31 Takeaways and questions Geometry, labels, signs. Ask about circuits and biased noise. Leave QR up.

42:35 script ends. Stop by 58:00.

Q&A / B1–B5 Backups Reed–Muller lengths, rank reduction, finite cases, dictionary with [CLM+26], the 47-qubit comparison.